

Software Engineering

Software Requirement Specifications, Project Plan, Draft Architecture and Design

E-commerce Website: Aesthetic Front-End and Functional Back-end

Software Requirements Specification (SRS)

1. Goal

The primary goal is to build a fully deploy-able, high-performance E-commerce platform that provides a seamless shopping experience for customers and robust management tools for administrators.

2. Key Deliverables

- Fully functional Client/Retail-facing web application.
- Comprehensive Biz Side (Admin) dashboard for operations.
- Secure Back end API and integrated database.

3. Functional Requirements (FR)

FR:3.1 *Client/Retail Side:*

FR: 3.1.1: Product Catalog: Search and filter functionality for items.

FR: 3.1.2: Shopping Cart: Add, remove, and update item quantities.

FR: 3.1.3: Checkout & Payment: Integration with electronic payment systems.

FR:3.2 *Biz Side (Admin)*

FR:3.2.1: Inventory Management: CRUD operations for products and categories.

FR:3.2.2: Order Fulfillment: View and update status of customer orders.

FR:3.2.3: Transaction Acknowledgment System: Generating Invoice, Receipt for clients and Logs for Admins. In-app notifications and mails notifications.

FR:3.2.4: User Analytics: Basic reporting on sales and site traffic.

4. Non-Functional Requirements (NFR)

NFR:4.1: Usability & Design: Clean, aesthetic, semantic UI with minimalist design

NFR:4.2: Performance: Fast page load times and optimized back-end processing.

NFR:4.3: Security: Ensure security of data and payment processings.

NFR:4.4: Scalability: N-tier architecture to support increasing user traffic.

Project Plan

5. Team and Responsibilities:

- **Design Team:**
 - Gargee Kakaty (*Lead*)
 - Kaustuva Kashyap
- **Front-End Development Team:**
 - Abhishek Das (*Lead*)
 - Gargee Kakaty
- **Back-end Development Team**
 - Ronit Choudhury (*Co- Dev*)
 - Abhishek Das (*Co-Dev*)
- **Documentation Team:**
 - Ronit Choudhury
 - Kaustuva Kashyap

6. Development Methodology: RAD Evolutionary Model

The combined use of the Rapid Application Development (RAD) and Evolutionary Model is ideal for the E-commerce project due to the need for speed, adaptability, and high user focus.

Why RAD and Evolutionary Model are a good fit:

- Rapid Delivery and Early Feedback (RAD)
- Handling Evolving Requirements (Evolutionary Model)
- Risk Mitigation through Iteration
- Focus on Usability (RAD)

How the approach will be implemented:

- **Time-boxing:** The project will be broken down into short, fixed-length phases . Each phase will focus on delivering a functional subset of the

requirements, such as a "Minimum Viable Shopping Experience" followed by an "Admin Management Module."

- **Prototyping and Tool-Based Construction:** The Development teams will leverage integrated development environments (IDEs) and potentially low-code tools for rapid construction of the Biz Side (Admin) dashboard and the secure Backend API, maximizing code reuse and speed.
- **Concurrent Component Development:** The Front-end and Back-end teams will work in parallel. While the Back end team (*Ronit Choudhury and Abhishek Das*) develops the core API for Inventory Management, the Front-end team (*Abhishek Das and Gargee Kakaty*) can simultaneously build the client-side UI based on mock data, integrating the components in the later stages of each evolutionary cycle.
- **Iterative Refinement:** Each iteration will result in a tested, working deliverable. For example, the first iteration might focus only on the Product Catalog and basic searching, while subsequent iterations add the Shopping Cart, Checkout, and then the Admin-facing modules.

7. Architecture Design Drafts:

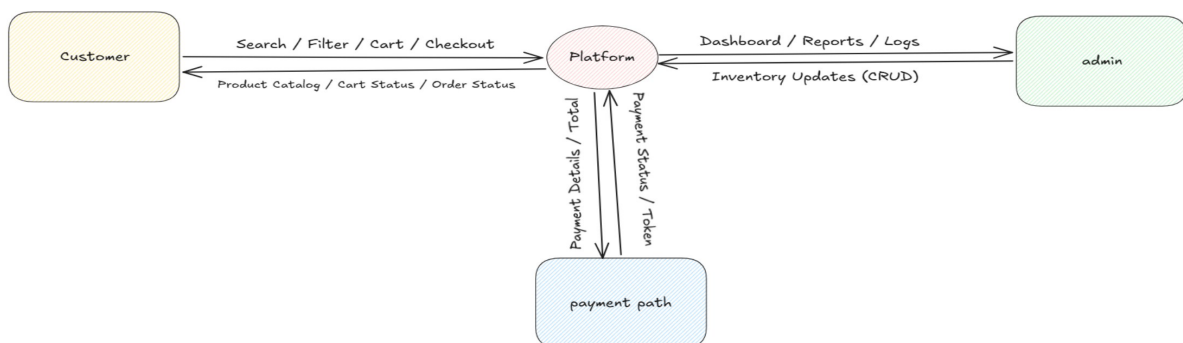
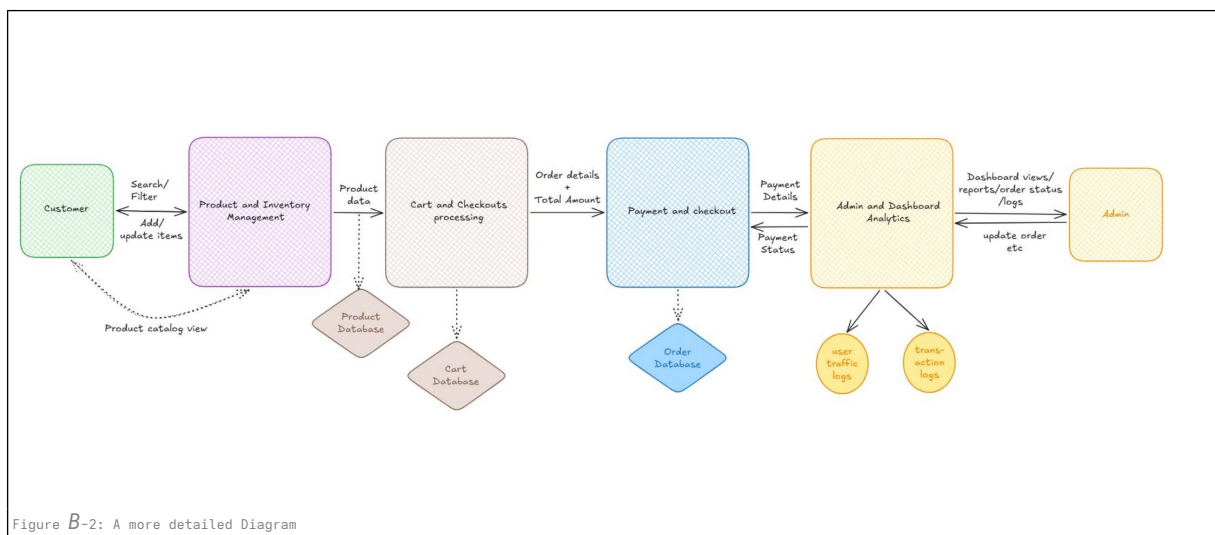


Figure A-1: High Level Architecture

High Level Architecture Design of the platform. The Main Components are:

- **Customer/Client** which supports the following high level functions:
 - o *Service Offered by the Interface:*
 - **Search:** Searching product catalogue
 - **Filter:** Filtering product catalogue by tags
 - **Cart:** Storing products for ordering for saving for future.
 - **Checkout** (includes orders and payment)
 - o *Servives Offered to the Interface:*
 - Product Catalogue Data
 - Cart Status
 - Order Status
- **A Main Platform Server:** Includes the Database and Main Logic
- **A Business/ Admin Client:** for Inventory Management/Operations, Reports, logs and Dashboard



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